Wavelength Division Multiplexing (WDM) (cont.)

Salwa El-Sabban

WDM

WDM Technology

Three specific groups

1-DWDM

2-CWDM

3-Cross Band

The difference between these groups of WDMs is in their channel spacing. This translates to system cost. The tighten the WDM channel spacing, the higher the WDM system price. This is primarily due to being more costly to manufacture WDMs with tightly spaced channels and to manufacture narrow spectra DWDM lasers that are temperature stabilized and will not drift outside of the narrower DWDM channel passband.

WDM

WDM Technology

DWDM is an ITU-T standard with channel spacing that is less than or equal to 1000 GHz (8 nm). Typically, DWDMs are available with channel spacing of 200 GHz and less, with the 200 GHz and 100 GHz being most common. With 200 GHz channel spacing a maximum of 22 channels are available in C band (two-fiber link) and with 100 GHz channel spacing a maximum of 44 channels are available in C band (two-fiber link).

CWDM is an ITU-T standard with channel spacing of 20 nm (approximately 2500 GHz).

Crossband WDMs have only two channels, one in the 1310 nm band (O band) and the other in the 1550 nm band (C band).

ITU: International Telecommunication Union
ITU-T: ITU Telecommunication standardization sector

WDM

DWDM

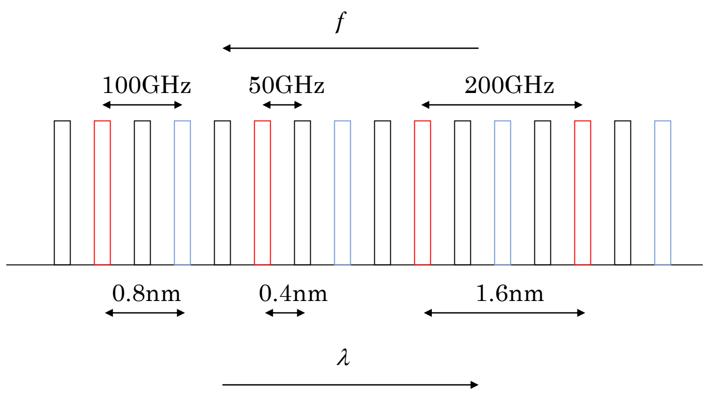
Dense wavelength division multiplexing (DWDM) is a WDM technology, where the channel spacing is less than or equal to 1000 GHz (8 nm), typically 200 GHz (1.6 nm) or less.

ITU-T has assigned a standard DWDM grid with channel center frequencies for 200, 100, 50, 25, and 12.5 GHz spaced DWDM systems.

Channel spacing of 25 GHz and less is referred to as Ultra-DWDM.

The grid is centered around 193.10 THz reference frequency.

DWDM System:



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WDM

DWDM

DWDM systems using 200 and 100 GHz channel spacing are very common. With 200 GHz channel spacing a maximum of 22 channels are possible in C band and using 100 GHz channel spacing 44 channels are possible.

Systems with higher channel counts are available with channel spacing of 50, 25, and 12.5 GHz using C, S, and L bands.

Maximum channel counts that are known currently exceed 200.

DWDM technology is ideal for networks that require high fiber channel counts and future expansion capacity for short- or long-haul links.

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DWDM

WDM

ITU-T 100 and 200 GHz DWDM Grid								Industry Typical Assignments‡			
	Band		Chan. No.†	Center Freq. (THz)	Center Wave. (nm)	100 GHz	200 GHz	8-chan.	16-chan.	32-chan.	44-chan.
			41	194.10	1544.53	x	x				x
			40	194.00	1545.32	x					x
			39	193.90	1546.12	x	x				x
			38	193.80	1546.92	x					x
			37	193.70	1547.72	x	x	x	x	x	x
			36	193.60	1548.51	x				x	x
			35	193.50	1549.32	x	x	x	x	x	x
			34	193.40	1550.12	x				x	x
			33	193.30	1550.92	x	x	x	x	x	x
			32	193.20	1551.72	x				x	x
			31	**193.10**	1552.52	x	x	x	x	x	x
			30	193.00	1553.33	x				x	x
EDFA Range	C	Red‡	29	192.90	1554.13	x	x	x	x	x	x
			28	192.80	1554.94	x				x	x
			27	192.70	1555.75	x	x	x	x	x	x
			26	192.60	1556.55	x				x	x
			25	192.50	1557.36	x	x	x	x	x	x
			23	192.30	1558.98	x	x	x	x	x	x

Red (upstream traffic) and Blue (downstream traffic) bands

Part of ITU-T 100 and 200 GHz DWDM Grid

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DWDM

Expanding Fiber Capacity Using Basic DWDMs

Following table estimates maximum fiber cable lengths using G.652 standard fiber with attenuation of 0.25 dB/km @ 1550 nm for various transceivers and basic (passive) DWDMs. DWDM insertion loss (IL) is shown for one unit and is multiplied by two for total span IL. Cable lengths greater than these can be achieved by deploying optical amplifiers and dispersion compensation modules (where required).

DWDM Transceiver	4-Channel IL = 2.0 dB	8-Channel IL = 3.5 dB	16-Channel IL = 4.0 dB	32-Channel IL = 5.0 dB
10 Gbps Budget is 23 dB	76 km	64 km	60 km	52 km
2.5 Gbps Budget is 26 dB	88 km	76 km	72 km	64 km
1 Gbps Budget is 26 dB	88 km	76 km	72 km	64 km

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DWDM

Expanding Fiber Capacity Using Basic DWDMs

Example Two dark fibers in a fiber cable are being leased by a company between its two offices. The fiber cable length $L = 50$ km. The dark fibers are currently connected to the company's 10 GigE switch, which is equipped with 1550 XFP transceivers, and provides high-speed intranet office communications. The company wants to expand its fiber link capacity without leasing additional fibers to meet current demands and future expansion requirements to at least eight 10 Gbps channels that can support various protocols including the current 10 GigE system, other GigE systems, SONET/SDH, and ATM systems. The solution needs to be highly reliable and be able to be installed quickly in one night with minimal disruption to the existing communications system. The measured dark fiber link loss on each fiber is 12.0 and 12.5 dB.

DWDM

Expanding Fiber Capacity Using Basic DWDMs

A good solution for this company is to deploy a basic (passive) DWDM system. It is protocol independent, rate independent, highly reliable and simple to install with typical down time of less than 30 minutes.

To determine the maximum number of channels this link can support, basic DWDM manufacturer's insertion loss specifications are obtained for 8-, 16-, and 32-channel DWDMs and are as follows:

8-channel IL: 3.5 dB
16-channel IL: 4.5 dB
32-channel IL: 7.0 dB

DWDM

Expanding Fiber Capacity Using Basic DWDMs

To determine the total link loss, following Equ. can be used. The DWDM insertion loss is multiplied by two because it is inserted at both ends of the fiber link.

$$\Gamma_T = 2 \times IL_{DWDM} + \Gamma_{fiber}$$

where
Γ_T = total link loss, dB
Γ_{fiber} = fiber loss, dB
IL_{DWDM} = DWDM insertion loss, dB

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DWDM

Expanding Fiber Capacity Using Basic DWDMs

Adding these DWDM units to the fiber link (worst-case fiber) would result in total link loss as follows:

8-channel IL plus fiber maximum link loss:
$$2 \times 3.5 + 12.5 = 19.5\text{dB}$$
16-channel IL plus fiber maximum link loss:
$$2 \times 4.5 + 12.5 = 21.5\text{dB}$$
32-channel IL plus fiber maximum link loss:
$$2 \times 7.0 + 12.5 = 26.5\text{dB}$$

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DWDM

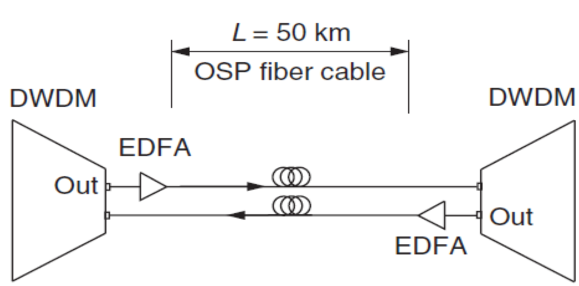
Expanding Fiber Capacity Using Basic DWDMs

Worst-case optical budget for transceivers is determined to have an optical budget of 23 dB that includes a 2 dB dispersion penalty. Therefore, it is clear that either the 8-channel or 16-channel DWDM system can be used since they are both below 23 dB.

If the company wanted to use the 32-channel system, all that is required, in this case, is to deploy optical amplifier EDFAs with a gain of 5 dB or more. They are placed after the DWDMs on the DWDM common out fiber at both ends of the link to compensate for the high DWDM insertion loss, as in fig. 1.

DWDM

Expanding Fiber Capacity Using Basic DWDMs



Basic 10 Gbps DWDM 50 km link with EDFAs

Fig. 1

DWDM

Expanding Fiber Capacity Using Basic DWDMs

-Dark fiber refers to unused fiber-optic cable. Sometimes companies lay more lines than what's needed in order to limit costs of having to do it again and again. The dark fiber can be leased to individuals or other companies who want to establish optical connections among their own locations.

-Fiber dispersion degrades the performance of optical communication systems by broadening optical pulses as they propagate inside the fiber.

-Dispersion Penalty: The result of Dispersion in which Pulses spread making it difficult for the Receiver to distinguish between ones and zeros. This results in a Loss of Receiver Sensitivity compared to a short Fiber and measured in dB.

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WDM

CWDM

-Coarse wavelength division multiplexing (CWDM) is a WDM technology where the wavelength spacing is 20 nm (approximately 2500 GHz as defined by ITU-T.

-Eighteen channels are defined in the ITU standard. The first two channels with center nominal wavelengths of 1271 and 1291 nm may exhibit high loss and may be unusable in some fiber types. Channels 1351 to 1451 nm occur near the 1383 nm fiber water peak where fiber attenuation is high, typically >2 db/km, and may also be unusable. However, this high loss region is flattened in newer "full spectrum" type fiber.

-Because of lower fiber attenuation at higher wavelengths and water peak attenuation, 8-channel CWDM multiplexers typically use channels 1471 to 1611 nm and 4-channel units use channels 1611 to 1551 nm.

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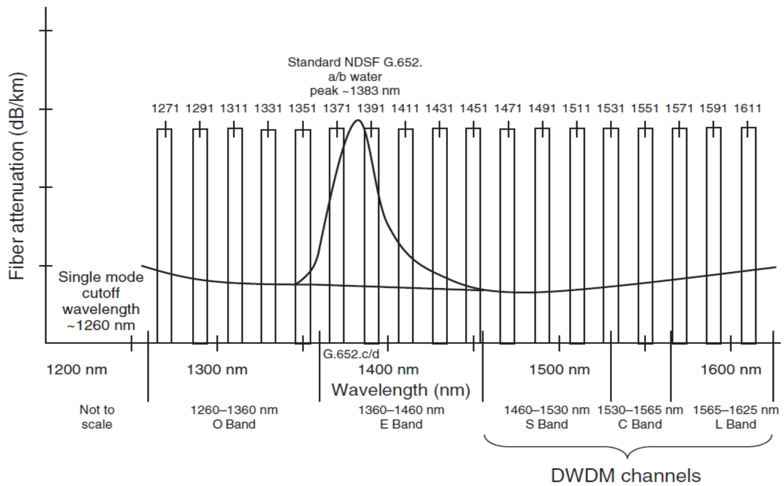
WDM

CWDM



CWDM fiber spectrum

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WDM

CWDM

CWDM transceivers are significantly less expensive than DWDM transceivers. This can be attributed to the factors listed below:

1. CWDM transceivers use simpler laser designs without elaborate temperature stabilization circuitry to keep the laser from drifting out of the DWDM narrow channel passband (60 GHz). The CWDM wider channel passband (1600 GHz) allows for laser drift without costly laser cooling and control circuitry.

2. Because of the large CWDM channel passband, transceivers have less stringent laser center wavelength and spectral width tolerances.

3. CWDM lasers use lower cost directly modulated DFB and VCSEL technology.

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WDM

CWDM

The CWDM lower cost technology is ideal for increasing fiber capacity up to 18 channels in metro fiber applications (<80 km).

It should be noted that non-CWDM transceivers that are labeled 1310 or 1550 nm will likely not work in CWDM channels 1311 or 1551 nm because of their large laser drift and wide manufacturing tolerances. CWDM passband is ± 6.5 nm.

Passband is approximately 13 nm (1600 GHz).

CWDM

Expanding Fiber Capacity Using Basic CWDMs

Advantages

*Both CWDM multiplexers and transceivers are less expensive than DWDM products.

*CWDM links are Ideal for shorter fiber spans.

*CWDM lasers consume less power than DWDM lasers (approximately 20% less24) because they use uncooled laser without any temperature stabilization circuitry.

CWDM

Expanding Fiber Capacity Using Basic CWDMs

Disadvantages

*Due to the wider spectral width of CWDM lasers, chromatic dispersion distance limitation is typically less than 80 km for most transceivers.

*Because CWDM band spans across the entire fiber spectrum, total fiber link loss and dispersion may change considerably between channels.

*For fiber that is not “full spectrum” (water peak at 1383 nm), channels 1351 to 1451 will suffer from high optical attenuation and it is likely unusable in 16-channel systems except possibly for very short links. Eight-channel CWDM systems are still possible (channels 1471 to 1611).

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CWDM

Expanding Fiber Capacity Using Basic CWDMs

Disadvantages

*Channels 1531, 1551, and 1571 use all of the C band spectrum that could be used to deploy a high channel count DWDM system.

*CWDM pluggable transceivers may have a worst maximum BER (typical 10^{-10}) compared to most DWDM transceivers’ maximum BER (typical 10^{-12}). This higher BER may not be acceptable for some applications.

*To compensate for fiber loss, EDFA amplifiers cannot be used with this technology because of the EDFA restricted passband. However, Raman amplifier technology may be effective in some applications.

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CWDM

Expanding Fiber Capacity Using Basic CWDMs

Important

Because of the very wide fiber spectrum that may be used in CWDM deployment, fiber optical loss can vary greatly between channels. Loss measurements should be taken at every channel when designing the link to ensure optical budgets are met.

CWDM

Expanding Fiber Capacity Using Basic CWDMs

Following table estimates maximum fiber cable lengths using G.652 standard fiber with attenuation of 0.37 dB/km @ 1311 nm, 2.5 dB/km @ 1391 nm, 0.25 dB/km @ 1551 nm, for various equipment optical budgets and channel counts using basic (passive) CWDMs and shown insertion losses (IL). Cable lengths greater than these may be achieved by deploying Raman optical amplifiers and dispersion compensation modules (where required). CWDM IL is shown for one unit and is multiplied by two for total span IL.

CWDM

Expanding Fiber Capacity Using Basic CWDMs

CWDM	4-Channel IL = 2.5 dB			8-Channel IL = 3.5 dB			16-Channel IL = 5.0 dB		
Fiber attenuation	1311 nm 0.37 dB/km	1391 nm 2.5 dB/km	1551 nm 0.25 dB/km	1311 nm 0.37 dB/km	1391 nm 2.5 dB/km	1551 nm 0.25 dB/km	1311 nm 0.37 dB/km	1391 nm 2.5 dB/km	1551 nm 0.25 dB/km
100 Mbps to 2.7 Gbps transceiver with budget of 25 dB	54 km	8 km	80 km	48 km	7 km	72 km	40 km	6 km	60 km

Basic Passive CWDM System Maximum Cable Length Estimate Using standard G.652 Fiber

CWDM

Expanding Fiber Capacity Using Basic CWDMs

Example Two dark fibers in a fiber cable are being leased by a company between its two offices. The fiber cable length $L = 40$ km and the cable installation occurred 15 years ago. The dark fibers are currently connected to the company's GigE switch, which is equipped with 1550 SFP transceiver, and provides high-speed intranet office communications. The company wants to expand its fiber link capacity without leasing additional fibers to meet current demands and future expansion requirements to at least eight WDM 2.5 Gbps channels that can support various protocols including the current GigE system, other GigE systems, SONET/SDH, and ATM systems. The solution needs to be highly reliable and be able to be installed quickly in one night with minimal disruption to the existing communications system. The measured dark fiber link loss at 1551 nm on each fiber is 10.0 and 9.5 dB

CWDM

Expanding Fiber Capacity Using Basic CWDMs

Because of the long length, 40 km, the water peak channels will be useless if a 16-channel CWDM system is deployed. For an 8-channel system, loss measurements should be conducted at each end of the 8-channel spectrum and the center channel.

Measurement at 1551 nm resulted in fiber loss of 10.0 dB. Additional measurements at 1471 and 1611 nm result in fiber link loss reading of 11.5 dB. Therefore we will design to this maximum fiber loss.

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CWDM

Expanding Fiber Capacity Using Basic CWDMs

To determine the maximum number of channels this link can support, manufacturer's insertion loss specifications are obtained for 8- and 16-channel CWDMs and are as follows:
8-channel IL: 3.5 dB
16-channel IL: 5.0 dB

To determine the total link loss, following Equ. can be used. The CWDM insertion loss is multiplied by two because it is inserted at both ends of the fiber link.

$$\Gamma_T = 2 \times IL_{CWDM} + \Gamma_{fiber}$$

where
Γ_T = total link loss, dB
Γ_{fiber} = fiber loss, dB
IL_{CWDM} = CWDM insertion loss, dB

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CWDM

Expanding Fiber Capacity Using Basic CWDMs

Adding these CWDM units to the fiber link (worst-case fiber) would result in total link loss as follows:

8-channel CWDM IL plus fiber maximum link loss:

$$2 \times 3.5 + 11.5 = 18.5\text{dB}$$

16-channel CWDM IL plus fiber maximum link loss:

$$2 \times 5 + 11.5 = 21.5\text{dB}$$

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CWDM

Expanding Fiber Capacity Using Basic CWDMs

CWDM pluggable transceivers that work with transmission rates of 100 Mbps to 2.7 Gbps are available with optical budgets of 25 dB. Maximum link loss for both 8-channel and 16-channel CWDM designs are within optical budget for wavelengths 1471 to 1611 nm, 8-channel systems.

The 16-channel CWDM can be considered for use to obtain two or four more channels not in the water peak, 1331, 1311, 1291, and 1271 nm. Measuring losses at these wavelengths results in a maximum loss of 14.0 dB. Using previous Equ. , we obtain maximum link loss at these wavelengths as follows:

$$2 \times 5 + 14 = 24\text{dB}$$

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CWDM

Expanding Fiber Capacity Using Basic CWDMs

This is within the pluggable transceiver's optical budget of 25 dB. Therefore, as long as the high attenuation channels are avoided, a 16-channel CWDM can be used to achieve twelve usable channels.

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CWDM

CWDM Capacity Expansion Using DWDMs

It is possible to cascade a DWDM multiplexer with a CWDM multiplexer system to increase link capacity. CWDM channel passband of ±6.5 nm can pass up to sixteen 100 GHz channels and eight 200 GHz channels. CWDM channel 1531 nm has a passband of 1524.62 to 1537.38 nm, CWDM channel 1551 nm has a passband of 1544.62 to 1557.38 nm, CWDM channel 1571 nm has a passband of 1564.62 to 1577.38 nm (assuming laser drift of ±0.12 nm).

These passbands fall within the DWDM S, C, and L bands, where DWDM lasers (SFP, Xenpaks, etc.) are available.

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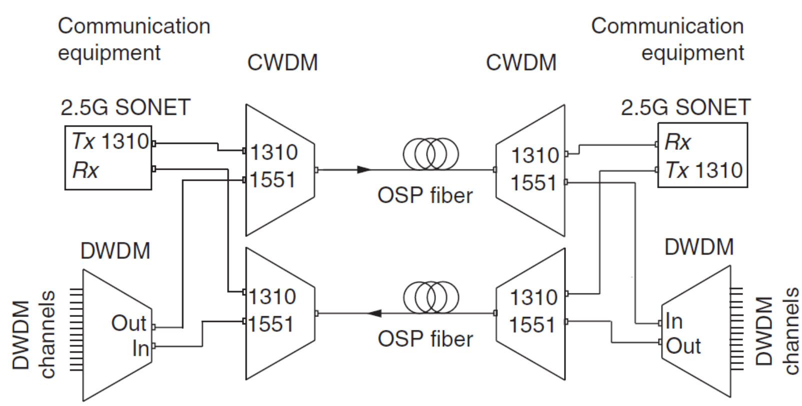
DWDM

CWDM Capacity Expansion Using DWDMs



DWDM multiplexers cascaded with CWDM multiplexers

CWDM

CWDM Capacity Expansion Using DWDMs

However, two concerns need to be addressed for such a configuration to be practical:

1. DWDM multiplexer insertion loss adds to the CWDM link loss decreasing the link optical budget and transmission distance.
2. Total DWDM aggregate output power can interfere with other CWDM channels.

CWDM

CWDM Capacity Expansion Using DWDMs

The first concern can be addressed by including DWDM insertion loss in total link loss calculations to ensure the equipment optical budget is not exceeded.

The second concern can cause interference on adjacent CWDM channels if the aggregate DWDM channel power is too high. Typical DWDM laser output power is approximately 0 dBm. With eight DWDM channels feeding into one CWDM channel, the total aggregate DWDM power into that one CWDM channel is 9 dBm minus the DWDM insertion loss. This power is significantly higher than the other CWDM channel typical launch powers of approximately 0 dBm. DWDM signal spillover to the other CWDM channels can decrease CWDM channel OSNR, thereby affecting BER performance.

CWDM

CWDM Capacity Expansion Using DWDMs

Calculations can be made to determine the amount of spillover to adjacent channels as shown in the following example.

Example An 8-channel DWDM multiplexer will be cascaded with an existing 16-channel CWDM multiplexer system. The existing 16-channel CWDM adjacent channel isolation specification is 28 dB. Insertion loss of the DWDMs and CWDMs are 3 dB each. Equipment CWDM laser launched power is 0 dBm and receiver sensitivity is –23 dBm @ 10^{-10} BER and OSNR 21 dB. Equipment DWDM laser launched power is 0 dBm and receiver sensitivity is –25 dBm @ 10^{-12} BER. Fiber total loss is 10 dB.

CWDM

CWDM Capacity Expansion Using DWDMs

CWDM channel total link loss is the total fiber loss plus two times the CWDM insertion loss, which calculates to 16 dB. DWDM channel total link loss is the total fiber loss, plus two times the CWDM insertion loss and plus two times the DWDM insertion loss, which calculates to 22 dB.

$\Gamma_{T-\mathrm{CWDM}} = 2 \times IL_{\mathrm{CWDM}} + \Gamma_{\mathrm{fiber}}$
$\Gamma_{T-\mathrm{CWDM}} = 16\mathrm{dB}$
$\Gamma_{T-\mathrm{DWDM}} = 2 \times IL_{\mathrm{CWDM}} + 2 \times IL_{\mathrm{DWDM}} + \Gamma_{\mathrm{fiber}}$
$\Gamma_{T-\mathrm{DWDM}} = 22dB$
where
$\Gamma_{T-\mathrm{DWDM}}$ = total DWDM channel link loss, dB
$\Gamma_{T-\mathrm{CWDM}}$ = total CWDM channel link loss, dB
IL_{DWDM} = DWDM insertion loss, dB
IL_{CWDM} = CWDM insertion loss, dB
Γ_{fiber} = fiber link loss, dB

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CWDM

CWDM Capacity Expansion Using DWDMs

$P_{Rx\text{-}\mathrm{CWDM}} = P_{Tx-\mathrm{CWDM}} - \Gamma_{T-\mathrm{CWDM}}$
$P_{Rx\text{-}\mathrm{CWDM}} = -16\ \mathrm{dBm}$
$P_{Rx\text{-}\mathrm{DWDM}} = P_{Tx-\mathrm{DWDM}} - \Gamma_{T-\mathrm{DWDM}}$
$P_{Rx\text{-}\mathrm{DWDM}} = -22\ \mathrm{dBm}$

where

$P_{Rx-\mathrm{DWDM}}$ = equipment receive power for a DWDM channel, dBm
$P_{Rx-\mathrm{CWDM}}$ = equipment receive power for a CWDM channel, dBm
$P_{Tx\text{-}\mathrm{DWDM}}$ = equipment DWDM laser transmit power
$P_{Tx\text{-}\mathrm{CWDM}}$ = equipment CWDM laser transmit power

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CWDM

CWDM Capacity Expansion Using DWDMs

With CWDM laser launch power of 0 dBm,
$P_{Rx-\text{CWDM}} = -16$ dBm.
CWDM signal maximum noise level is at −37 dBm
Maximum noise level= $P_{Rx-\text{CWDM}}$ −OSNR= -16-21= -37 dBm for BER of 10^{-10}.

The noise is mainly from the interference from adjacent channel.

Assuming interference for a CWDM channel will be received from both adjacent channels (from two sides instead of from 1 side), the signal noise level is decreased by 3 dB to −40 dBm.

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CWDM

CWDM Capacity Expansion Using DWDMs

With DWDM laser launch power of 0 dBm (1mW),
total aggregate power (8 channels) =1mWx8 = +9.0 dBm (8 mW).

Power out of the DWDM =9-3=6 dBm
(DWDM IL subtracted) and into the CWDM channel.

At the other end of the CWDM fiber link,
the CWDM channel signal power is now =6-16=−10 dBm (DWDM out minus CWDM link loss).

With a specified 28 dB CWDM channel isolation the adjacent channel noise maximum due to this aggregate DWDM signal = -10 −28= -38 dBm.
This is greater than the CWDM signal noise at −40 dBm and a degrade in adjacent CWDM OSNR can be expected.

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CWDM

CWDM Capacity Expansion Using DWDMs

If three DWDM channels are removed, the total aggregate power for five signals is 6.99 dBm (5 mW).

Power out of the DWDM is 3.99 dBm. At the other end of the CWDM fiber link the CWDM channel signal power is now −12.0 dBm (DWDM output minus CWDM link loss).

With a specified 28 dB CWDM isolation, the adjacent channel noise maximum is now −40 dBm. This is now at the specification noise limit for equipment OSNR −21 dB @ BER 10^{-10}.

The DWDM channel receive power is −22 dBm, which is determined from equipment DWDM laser signal launch minus DWDM channel link loss. This is within the equipment sensitivity limit of −25 dBm @ BER 10^{-12}.

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CWDM

CWDM Capacity Expansion Using DWDMs

Instead of decreasing the DWDM channel count, another option would be to decrease each DWDM channel launch power equally.

If we decrease each DWDM signal launch power by 2 dB, from 0 to −2 dBm, the same result can be achieved.

The total aggregate power for eight signals is +7.0 dBm. Power out of the DWDM is 4.0 dBm.

At the other end of the CWDM fiber link the CWDM channel signal power is now −12.0 dBm (DWDM output minus CWDM link loss).

With a specified 28 dB CWDM isolation, the adjacent channel noise maximum is now −40 dBm.

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CWDM

CWDM Capacity Expansion Using DWDMs

This is now at the specification noise limit for equipment OSNR −21 dB @ BER 10^{-10}.

The DWDM channel receive power is −24 dBm, which is determined from equipment DWDM laser signal launch minus DWDM channel link loss. This is still within the equipment sensitivity limit of −25 dBm @ BER 10^{-12}.

WDM

Cross-Band WDM

Cross-band WDMs combine two channels from different bands into one fiber. One channel has an O-band passband (1260 to 1360 nm) and the other channel has a C-band passband (1530 to 1565 nm).

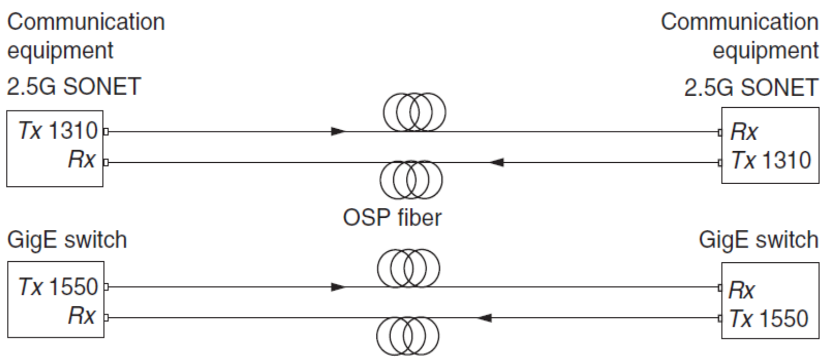
Lasers with any wavelength in these two bands can be used with these WDMs. The lasers do not need to meet any specific standard (such as ITU) or spectral width because the WDM channel passband is very wide, the entire band. Therefore, any laser with wavelength in O or C bands can be used to double the fiber's capacity.

Following figure shows a 2.5G SONET system with 1310 nm lasers and a GigE switch with 1550 nm lasers that normally would require four fibers for communication.

Cross-Band WDM

WDM

This figure shows a 2.5G SONET system with 1310 nm lasers and a GigE switch with 1550 nm lasers that normally would require four fibers for communication



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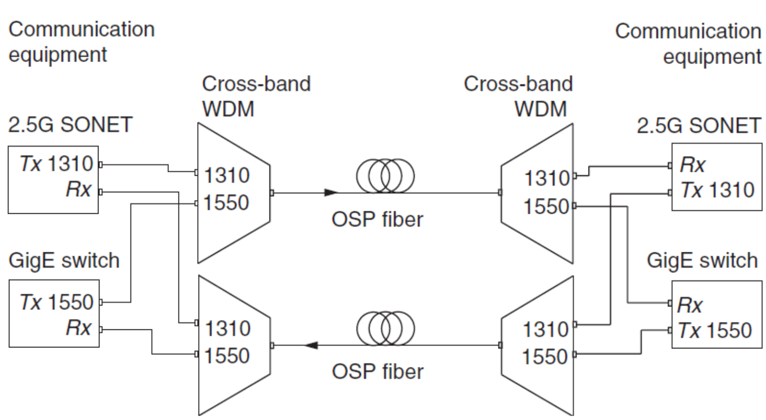
Cross-Band WDM

WDM

By adding four cross-band WDMs as shown in the following figures, the fiber count is reduced to two fibers using the same equipment lasers. The fiber capacity is double with these cross-band WDM units



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Cross-band WDM application

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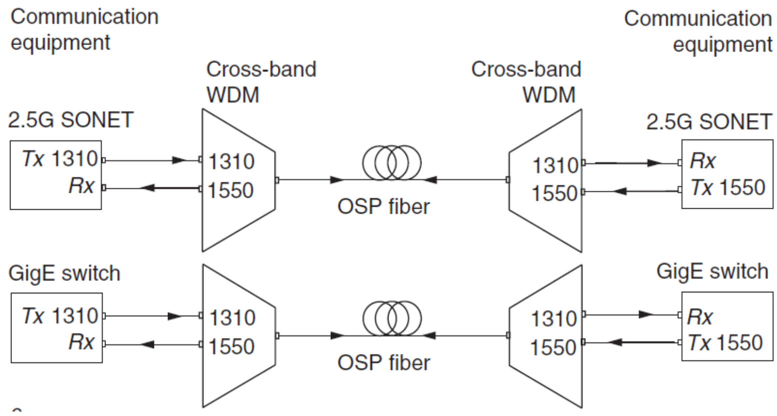
Cross-Band WDM

WDM



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Cross-band WDM application

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Cross-Band WDM

WDM

Two configurations are possible using cross-band WDMs:

Figure a shows the most common where unidirectional cross-band WDMs are deployed and each system uses wavelengths in each band.

In figure b, universal cross-band WDMs are deployed and each system uses one dedicated fiber for communications. Lasers at both ends of the same system must use different band lasers.

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WDM

Cross-Band WDM

Cross-band WDMs can also be combined with DWDMs to allow O-band transmission in a DWDM system. Following figure shows a DWDM system combined with a 1310 nm SONET system using cross-band WDMs. Since the cross-band WDM passband is the entire C band, the DWDM can have channels only in C band. Note that the total link loss includes the sum of the DWDM and WDM insertion losses.

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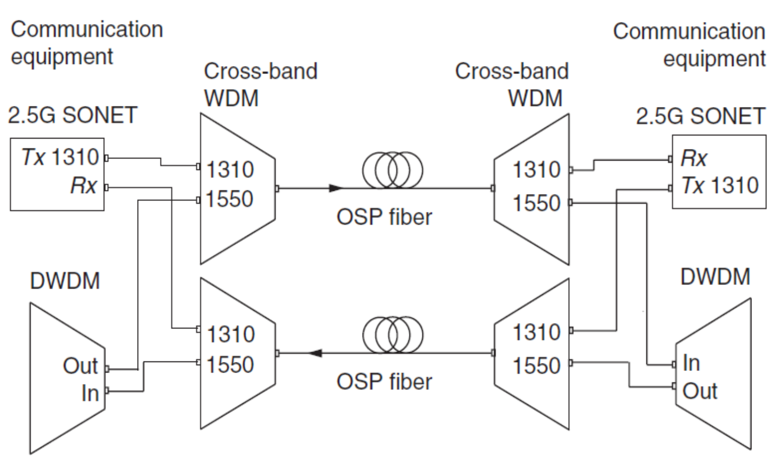
WDM

Cross-Band WDM



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Cross-band WDM and DWDM application

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WDM

WDM Specifications

The following WDM specifications are commonly used to define operating parameters.

Insertion Loss (IL)

This is the optical power loss measured between a WDM channel input and output ports. The measurement unit is dB. Maximum insertion loss for any WDM channel is provided in specifications.

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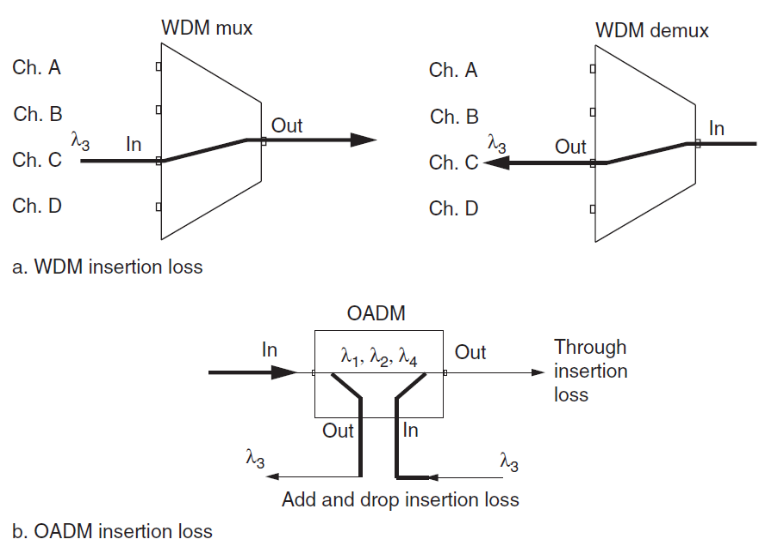
WDM

WDM Specifications



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WDM insertion loss diagram

WDM

WDM Specifications

Insertion Loss (IL)

Maximum insertion loss for FBG and TFF WDMs can also be specified for a mux/demux matched pair. This is the total loss for both multiplexer and demultiplexer units in the fiber link.

Internally, the optical WDM signals propagate from one channel filter to the other in a sort of series fashion. The signal that matches the channel's wavelength is dropped or else it continues to the next channel filter. Hence, filters closer to the aggregate (common) fiber end have a lower insertion loss than filters further from it.

Filter channels in matched multiplexer and demultiplexer pairs are sequenced in opposite directions from aggregate fiber such that the combined loss for each channel is more uniform and less than the combined loss of the two highest loss channels. This method provides a lower maximum insertion loss for the pair.

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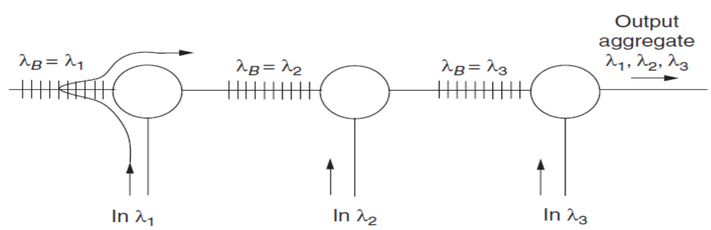
WDM

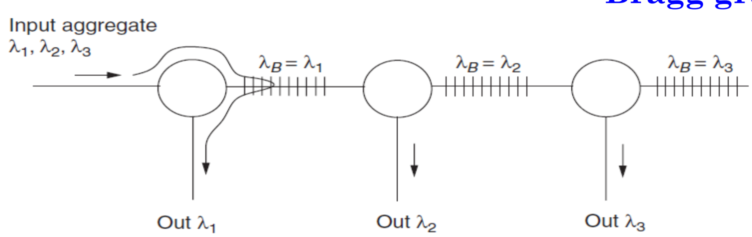
WDM Specifications

Insertion Loss (IL)



b. Three-channel WDM mux

Bragg gratings



c. Three-channel WDM demux

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WDM

WDM Specifications

Insertion Loss (IL)

For OADMs, the insertion loss is divided into through loss and channel add/drop loss. The OADM through loss is measured between the two common ports. The OADM channel add/drop loss is measured from the common port and a drop port.

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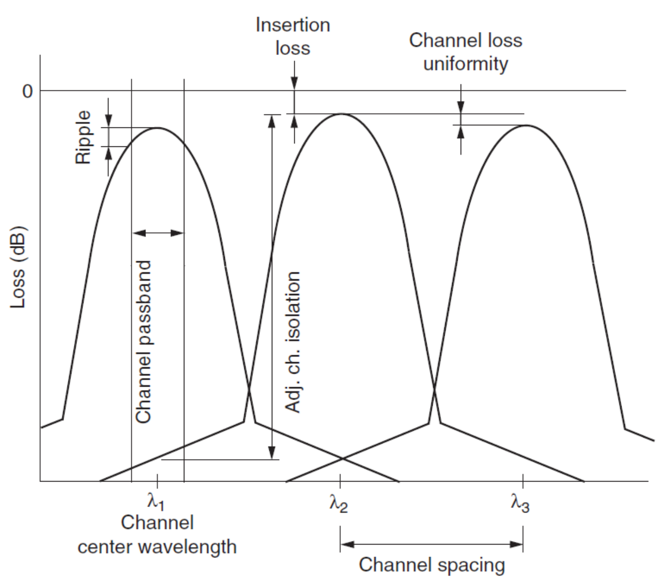
WDM

WDM Specifications



WDM Filter Plot

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WDM

WDM Specifications

Channel Spacing

This is the ITU-T WDM channel spacing measured between center wavelengths of adjacent channels. DWDM values are 200, 100, 50, and 25 GHz. CWDM channel spacing is 20 nm.

Channel Loss Uniformity

This is the maximum insertion loss difference between all channels. Unit of measure is dB.

Channel Wavelength

This is the center wavelength of each channel. Unit of measure is nm.

WDM

WDM Specifications

Channel Passband

This is the WDM channel filter passband measured 0.5 or 3 dB down from filter center. Unit of measure is nm or GHz. Passband value varies among manufacturers and with channel spacing.
Typically values for a 0.5 dB down passband are 0.5 nm for 200 GHz spaced channels, 0.22 nm for 100 GHz spaced channels, and 0.1 nm for 50 GHz spaced channels.

Channel (Passband) Ripple or Flatness

This is the peak-to-peak difference in insertion loss in a channel’s passband. Unit of measure is dB. Typically, value is less than 0.5 dB.

WDM

WDM Specifications

Adjacent Channel Isolation (Demultiplexer)

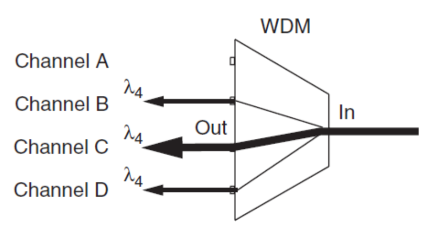
This is the maximum optical power leakage to adjacent channels referenced to launch channel power. Unit of measure is dB. A value greater than 25 dB is common.

Nonadjacent Channel Isolation (Demultiplexer)

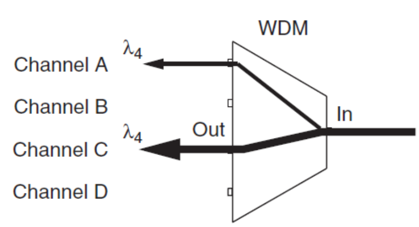
This is the maximum optical power leakage to nonadjacent WDM channels referenced to the launch channel power. Unit of measure is dB. A value greater than 45 dB is common.

WDM

WDM Specifications



a. Adjacent channel isolation



b. Nonadjacent channel isolation

WDM

WDM Specifications

Directivity (Multiplexer)

This is the maximum channel power reflected back into other WDM channels referenced to the launched channel power. Unit of measure is dB. A value greater than 50 dB is common.

Return Loss

This is the maximum optical power reflected back onto itself in the same channel referenced to the launched power. Unit of measure is dB. A value greater than 45 dB is common.

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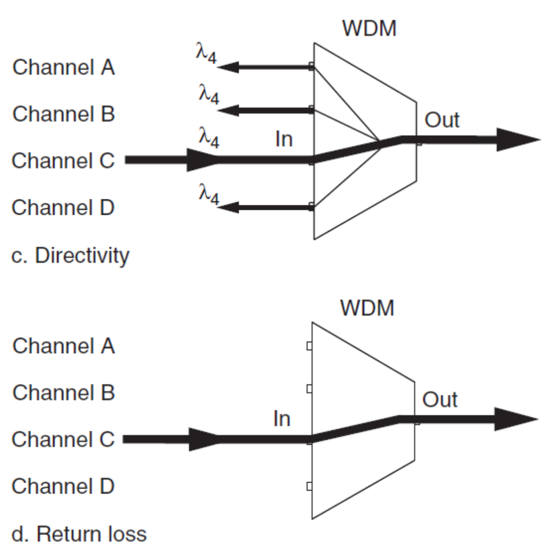
WDM

WDM Specifications



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